Formulas at Presentation Size

A short deck of large equations

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The Pythagorean Identity

$$\sin^2\theta + \cos^2\theta = 1$$

The Product Rule

$$(fg)' = f'g + fg'$$

Integration by Parts

$$\int u\,dv = uv - \int v\,du$$

Sum of an Arithmetic Series

$$S_n = \frac{n}{2}(a_1 + a_n)$$

Slopes and Rates

- The slope of a line is $\frac{y_2 - y_1}{x_2 - x_1}$
- Average speed is $\frac{\Delta d}{\Delta t}$ over the interval
- A gradient of $\frac{3}{4}$ rises three units for every four across

The Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Standard Error of the Mean

$$SE = \frac{s}{\sqrt{n}}$$

Kinetic Energy

$$K = \frac{1}{2}mv^2$$

Coulomb's Law

$$F = k \frac{q_1 q_2}{r^2}$$

Fractions in Practice

- A pump fills $\frac{1}{15}$ of the pool each hour
- Two pumps together fill $\frac{1}{15} + \frac{1}{20}$ per hour
- The probability of a head is $\frac{1}{2}$
- A quarter of a quarter is $\frac{1}{16}$

Radioactive Decay

$$N(t) = N_0 \, e^{-\lambda t}$$

The pH Scale

$$pH = -\log_{10}[H^+]$$

An Equilibrium Constant

$$K = \frac{[C]^c \ [D]^d}{[A]^a \ [B]^b}$$

Burning Butane

$$2\,C_4H_{10}(g) + 13\,O_2(g) \longrightarrow 8\,CO_2(g) + 10\,H_2O(l)$$

Averages

- The mean of n values is $\frac{x_1 + x_2 + \cdots + x_n}{n}$
- The variance divides by $n - 1$, not by n
- A weight of $\frac{2}{5}$ means forty percent

Beta Decay of Strontium

$$^{90}_{38}\mathrm{Sr} \longrightarrow {}^{90}_{39}Y + {}^{0}_{-1}e + \bar{\nu}_e$$

Inverse of a Two by Two Matrix

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

The Sign Function

$$\operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

A Determinant of Order Three

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Ratios and Proportions

- Equal ratios satisfy $\frac{a}{b} = \frac{c}{d}$
- Cross-multiplying gives $ad = bc$
- A scale of $\frac{1}{50}$ shrinks every length fifty times

Two Equations Together

$$\begin{cases} x + y = 7 \\ x - y = 1 \end{cases}$$